SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Laksha. M.P (192426030 2) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	5
Experimental Design	30	5
Use of Tools	20	5
Analysis of Results	15	5
Documentation	10	4

Total Marks 24 **/30**

Signature of the Faculty

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretatio n		
Documentati on	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentati on with necessary details	Basic documentati on with missing details	Poor or incomplete documentati on



ARTIFICIAL INTELLIGENCE

for Network Traffic Management

Smarter routing • Real-time monitoring • Reliable networks

1924260302 | LAKSHA.M P

Guided by: R. SHANKAR

Introduction & Need

Network traffic management monitors and controls data moving across devices, applications and the internet. AI helps analyze traffic patterns and respond to changing conditions.

Growing traffic

More users, connected devices, streaming and cloud services increase network demand.

Common problems

Congestion, latency, packet loss and sudden traffic spikes affect performance.

Why AI?

AI can find patterns quickly and support prediction, automation and better decisions.

Goal: provide a faster, more reliable and efficient network experience.

What Is AI-Based Traffic Management?

It uses AI to study network data, predict traffic conditions and recommend or automate actions that improve data flow.



The process repeats as new measurements arrive. Human oversight remains important.

How AI Works & Techniques Used

Collect data

Bandwidth use, delay, packet loss, flow records and device logs.

Prepare data

Clean it and convert it into useful measurements.

Analyze traffic

Find patterns, forecast demand and detect unusual behavior.

Take action

Recommend routes, adjust priority or alert an operator.

Learn from feedback

Measure results and refine models or policies.

Machine learning

Learns from past traffic patterns.

Anomaly detection

Flags unusual network activity.

Traffic forecasting

Estimates future network demand.

Applications of AI

Smart routing

Choose paths using congestion, delay and link health.

Bandwidth allocation

Share capacity based on demand and priority.

Congestion prediction

Identify likely overload before performance drops.

Quality of Service

Prioritize voice, video and critical applications.

Anomaly detection

Spot unusual spikes or possible attacks.

Predictive maintenance

Identify possible device or link problems early.

Example: A Busy Campus Network

Scenario: students begin streaming videos and attending online classes at the same time.

1. Detect

AI notices increasing bandwidth use and delay at a wireless access point.

2. Respond

The controller balances suitable traffic or applies fair priority rules.

3. Check

The system measures delay and packet loss to see whether performance improves.

Expected benefit: fewer bottlenecks and a more consistent experience, depending on network capacity and design.

Benefits of AI-Based Management

Faster response

Detects changing conditions quickly.

Improved performance

Can reduce congestion and service delays.

Efficient resource use

Makes better use of bandwidth and routes.

Predictive operations

Helps identify risks before failures occur.

Scalability

Supports monitoring of large networks.

Less manual effort

Automates routine monitoring and recommendations.

Challenges, Security & Privacy

Data quality

Inaccurate data can produce poor decisions.

Privacy

Traffic analysis must follow data protection rules.

Security

Attackers may manipulate data or automated actions.

Complexity and cost

Tools and integration need skilled staff.

Over-automation

Incorrect actions can disrupt important services.

Protect data

Limit access and collect only necessary information.

Human oversight

Keep approval, rollback and escalation procedures.

Test regularly

Check models and monitor false alarms.

Future Scope & Conclusion

5G and future networks

AI can help manage high-speed, complex mobile traffic.

Edge AI

Analyze data closer to users for faster local decisions.

Self-optimizing networks

Systems may adjust settings within safe policies.

Conclusion

AI can help networks analyze traffic, anticipate congestion and use resources efficiently. Its best results depend on reliable data, strong security, careful testing and human supervision.

Key message: smarter networks need intelligent automation and responsible control.

THANK YOU!


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TRAFFIC MANAGEMENT
GPS Map Camera
Karanaipuducheri, Tamil Nadu, India
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